CHARACTERIZATION AND PREDICTION OF EARLY REVIEWERS FOR PRODUCT MARKETING IN E-COMMERCE PLATFORMS

A Project Report

Submitted in the Partial Fulfillment of the Requirements for

The Award of the degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

(DATA SCIENCE)

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CERTIFICATE

This is to certify that this project work entitled **" CHARACTERIZATION AND PREDICTION OF EARLY REVIEWERS FOR PRODUCT MARKETING IN E-COMMERCE PLATFORMS "** is a bonafide record of project work done **K. SAHITYA (228B1A4419), B. ANUSHA (228B1A4403), G. DEVIKA (228B1A4409)** in partial fulfillment of the requirements for **IV Year - II Sem** of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)** from **JNTUK,** during the period of **2025 - 26** is a record of bonafide work carried out by him/her under our esteemed guidance and supervision.

PROJECT GUIDE **HEAD OF THE DEPARTMENT**

EXTERNAL EXAMINER

DECLARATION

We hereby declare that the work is being presented in this dissertation entitled **"Characterization and Prediction of Early Reviewers for Product Marketing in E-Commerce Platforms"** submitted towards the partial fulfillment of requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Data Science)** work carried out under the supervision of **Mr. G. Gopi** Assistant professor, **Rise Krishna Sai Gandhi Group of Institutions, Valluru.** The results embody in this dissertation report have not been submitted by me for the award of any other degree. Furthermore, the technical details furnished in various chapters of this report are purely relevant to the above project and there is no deviation from the theoretical point of view for design, development and implementation.

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ABSTRACT

Online customer reviews play a major role in shaping buyer opinion and influencing product success in e-commerce platforms. Among all reviewers, early reviewers are especially important because their opinions appear during the initial stage of product exposure and can affect future customer interest and product visibility. This paper presents a machine learning-based approach to analyze the characteristics of early reviewers and predict which users are likely to post reviews early. The proposed work uses historical review data collected from e-commerce datasets and examines behavioral and temporal factors such as review timing, reviewer activity, rating patterns, and helpfulness-related information. Based on these features, classification models are trained to distinguish early reviewers from regular reviewers. The study also highlights the relationship between early reviewing behavior and product popularity. Experimental evaluation shows that the proposed approach can effectively support early reviewer identification and provide useful insights for product marketing and decision-making in online marketplaces.

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1. INTRODUCTION

The rapid growth of e-commerce platforms has changed the way customers search, evaluate, and purchase products. Online customer reviews have become an important factor influencing buying decisions, as they provide insights into product quality and user satisfaction.

Early reviews play a crucial role in determining the success of a product. Positive feedback in the initial stage increases product visibility and trust, while negative feedback may reduce customer interest. Therefore, early reviewers have a significant impact on shaping product reputation in online marketplaces.

However, identifying early reviewers in advance is a challenging task due to diverse user behavior. Most existing systems focus only on analyzing reviews after they are posted, rather than predicting which users are likely to review early.

To address this issue, this project proposes a system that analyzes user behavior and review patterns to identify early reviewers. The system also performs sentiment analysis on reviews, helping in understanding customer opinions and improving decision-making in e-commerce platforms.

1.1 MOTIVATION

The motivation behind this project is to improve product marketing strategies in e-commerce platforms by identifying and predicting early reviewers who can influence product visibility, customer perception, and purchase decisions during the initial stage of a product launch. By using machine learning techniques to analyze historical review data, reviewer activity, rating behaviour, and temporal patterns, the project aims to provide an intelligent and data-driven approach for understanding early reviewer behaviour and supporting proactive marketing decisions. It seeks to address the limitation of existing systems that analyze reviews only after they are posted, thereby helping sellers and marketers engage potential early reviewers in advance. Ultimately, the goal is to enhance product promotion, strengthen customer trust, and support better decision-making in online marketplaces.

1.2 PROBLEM DEFINITION

In e-commerce platforms, customer reviews play an important role in influencing purchasing decisions and building trust among users. However, newly launched products often lack sufficient reviews during the early stages, making it difficult for customers to evaluate their quality and reliability. Early reviewers can help address this issue by providing initial feedback, but identifying users who are likely to review products early is a challenging task. Most existing systems analyze reviews only after they are posted and do not focus on predicting early reviewers in advance. Therefore, there is a need to develop an intelligent system that can analyze user behaviour and predict potential early reviewers, helping businesses generate timely feedback, improve product visibility, and support better marketing strategies in online marketplaces.

1.3 OBJECTIVE OF THE PROJECT

The main objective of this project is to develop a system that can identify and predict early reviewers in e-commerce platforms using machine learning techniques. The project aims to analyze historical review data, user behaviour, rating patterns, and review timing to determine which users are likely to review newly launched products early. Another objective is to understand the role of early reviewers in improving product visibility and influencing customer decisions. In addition, the system aims to help sellers and marketers improve product promotion strategies and encourage early feedback. Overall, the project focuses on supporting intelligent review analysis and better marketing decisions in online marketplaces.

2. LITERATURE SURVEY

1. Title: “The Effect of Word of Mouth on Sales: Online Book Reviews” Author: Judith A. Chevalier and Dina

Description: This paper studies how online customer reviews influence product sales and customer purchasing behaviour in e-commerce platforms. The authors explain that reviews play an important role in shaping product demand and market performance. Positive reviews can improve product visibility and attract more buyers, while negative reviews may reduce customer interest. The study highlights the value of customer feedback in building trust and influencing the success of products in online marketplaces.

2. Title: “Mining and Summarizing Customer Reviews”

Author: Minqing Hu and Bing Liu

Description: This paper focuses on extracting useful information from customer reviews and summarizing it effectively. The authors propose methods to identify important product features mentioned in reviews and determine whether opinions about those features are positive or negative. The work helps customers understand product strengths and weaknesses in a short time. It also shows how review mining can support better product analysis and improve decision-making in e-commerce systems.

3.Title: “Sentiment Analysis and Opinion Mining”

Author: Bing Liu

Description: This study explains how sentiment analysis techniques can be used to identify customer opinions expressed in online reviews. It describes methods for classifying reviews into positive, negative, and neutral sentiments, which helps businesses understand customer satisfaction and preferences. The paper is important in the field of e-commerce because it provides a foundation for analyzing large volumes of review text. It also supports the use of review-based insights for improving products and services.

4. Title: "Hidden Factors and Hidden Topics: Understanding Rating Dimensions with Review Text"

Author: Julian McAuley and Jure Leskovec

Description: This paper examines the relationship between review text and rating behavior in online platforms. The authors combine textual review information with rating data to identify hidden factors that influence customer preferences and product performance. The study shows that review data contains valuable behavioral patterns that can be used for analysis and prediction. This work is useful for understanding how user-generated review data can support intelligent systems in e-commerce environments.

5. Title: "Explaining and Predicting Online Review Helpfulness: The Role of Content and Reviewer-Related Signals"

Author: Michael Siering, Jan Muntermann, and Bala Rajagopalan

Description: This paper investigates the factors that make online reviews helpful to other customers. It considers both review content and reviewer-related signals such as credibility, activity, and behaviour. The authors show that reviewer characteristics strongly influence the usefulness of reviews. This study is relevant to the proposed project because helpful reviews, especially in the early stages of product release, can affect customer trust, buying decisions, and product perception.

6. Title: "The Effect of Online Consumer Reviews on New Product Sales"

Author: Geng Cui, Hon-Kwong Lui, and Xiaoning Guo

Description: This paper studies the impact of online consumer reviews on the sales performance of newly launched products. The authors found that customer reviews significantly affect product visibility and market success, particularly during the initial launch stage. The research highlights that early customer feedback can influence future sales and customer interest. This supports the importance of analyzing reviewer behaviour and understanding the role of early reviews in e-commerce product marketing.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In existing e-commerce platforms, online customer reviews play an important role in helping customers evaluate products and make purchasing decisions. Most e-commerce websites allow users to post ratings and reviews after purchasing and using a product. These reviews are then displayed on the product page so that other customers can read the feedback and understand the quality, performance, and reliability of the product. Businesses also analyze these reviews to understand customer satisfaction and identify areas where products or services can be improved.

The existing systems mainly focus on collecting and analyzing customer feedback after the reviews are posted. Different data analysis techniques are used to evaluate customer opinions, detect product trends, and measure overall product popularity. Some systems also apply sentiment analysis to classify reviews as positive, negative, or neutral in order to understand customer attitudes toward specific products. This information is often used by companies to improve product design, marketing strategies, and customer support services.

However, these existing systems operate in a reactive manner because they analyze reviews only after they have already been submitted by users. They do not focus on predicting which users are likely to provide early reviews for newly launched products. Early reviews are extremely important because they strongly influence the initial perception of a product and can affect its success in the market. When a product is newly introduced, customers may hesitate to purchase it if there are no reviews available. As a result, the absence of early feedback can slow down product adoption and reduce customer confidence.

In addition, existing systems generally treat all reviewers equally without analyzing the behavioral patterns of users who frequently provide early feedback. These systems do not attempt to identify influential users or early reviewers who could contribute valuable opinions during the initial stage of product marketing. Because of this limitation, businesses miss opportunities to encourage early feedback and promote products more effectively.

Challenges in Data Collection

Large Data Volume: E-commerce platforms generate a huge amount of review data every day, making it difficult to collect and manage the data efficiently.

Data Inconsistency: Review data may come from different sources and formats, which leads to inconsistencies in the collected dataset.

Fake or Spam Reviews: Some users may post fake or misleading reviews, which affects the quality and reliability of the collected data.

Privacy and Access Restrictions: Many platforms have strict privacy policies and data access limitations, making it difficult to obtain complete datasets.

Data Preprocessing in Existing Systems

Data Cleaning: Duplicate records, irrelevant information, and noisy data are removed from the dataset to improve data quality.

Handling Missing Values: Missing or incomplete data entries are identified and handled using appropriate techniques.

Data Transformation: Unstructured text reviews are converted into structured formats that can be used for analysis.

Feature Extraction: Important features such as review ratings, review time, and user activity patterns are extracted from the dataset.

Challenges in Data Preprocessing

Unstructured Data: Customer reviews are mostly written in natural language, which makes them difficult to process directly.

Noise in Data: Reviews may contain irrelevant content, spelling mistakes, or unnecessary information.

Spam Detection Difficulty: Identifying and removing fake or spam reviews is a complex task.

High Processing Time: Processing large datasets requires more time and computational resources.

Data Splitting in Existing Systems

Training Dataset: A portion of the dataset is used to train the predictive model.

Testing Dataset: Another portion of the dataset is used to evaluate the performance of the trained model.

Validation Dataset: Sometimes a validation dataset is used to fine-tune the model and improve accuracy.

Challenges in Data Splitting

Data Imbalance: If the dataset is not balanced, the predictive model may produce biased results.

Improper Distribution: Incorrect distribution of data between training and testing sets can affect model performance.

Loss of Important Patterns: If splitting is not done carefully, important patterns in the data may be lost.

Disadvantages

Reactive Analysis: Existing systems mainly analyze reviews after they are posted rather than predicting early reviewers.

Limited Prediction Capability: They do not effectively identify users who are likely to review products early.

Data Quality Issues: Presence of incomplete or spam reviews reduces the reliability of analysis.

Inefficient Marketing Support: The system cannot support proactive product marketing strategies during the early stage of product launch.

3.2 PROPOSED SYSTEM

The proposed system aims to identify and predict early reviewers in e-commerce platforms by analyzing user behavior, review patterns, and rating characteristics using advanced data analysis techniques. Unlike traditional systems that only analyze reviews after they are posted, the proposed system focuses on predicting which users are more likely to provide reviews in the early stages of a product's release. Early reviewers play an important role in shaping the initial perception of a product and influencing the purchasing decisions of other customers. By identifying these users in advance, businesses can encourage early feedback, improve product visibility, and strengthen product marketing strategies.

The proposed system integrates machine learning and natural language processing techniques to analyze large amounts of review data collected from e-commerce platforms. The system studies historical data such as user activity levels, frequency of reviews, rating patterns, and review timestamps to understand reviewer behavior. Based on these patterns, the machine learning model is trained to recognize users who have a higher probability of reviewing products early. This predictive approach allows businesses to proactively engage potential early reviewers rather than waiting for reviews to appear naturally.

In addition, the system uses natural language processing methods to analyze the textual content of reviews. By examining the sentiment, keywords, and opinions expressed in reviews, the system can gain deeper insights into customer feedback and product perception. This information not only improves the prediction of early reviewers but also helps businesses understand customer preferences and product performance.

The proposed system also supports personalized recommendations by analyzing user preferences and past interactions. Based on the review history and browsing behavior of users, the system can recommend relevant products that match their interests. At the same time, the system includes mechanisms to detect suspicious or spam review activities, which helps maintain the reliability and authenticity of customer feedback.

Overall, the proposed system provides a more intelligent and proactive approach to review analysis in e-commerce platforms. By predicting early reviewers and analyzing customer feedback effectively, the system helps businesses improve product marketing strategies, increase customer trust, and enhance the overall online shopping experience.

Data Collection

User Data Collection: The system collects information about users such as user IDs, activity history, and reviewing behavior.

Product Information Collection: Product-related details such as product ID, category, and product launch time are collected for analysis.

Review and Rating Data: Customer reviews, ratings, and timestamps are collected to understand review patterns and timing.

Historical Dataset: Previous review data is gathered to analyze past reviewer behavior and identify early reviewer patterns.

Data Preprocessing

Data Cleaning: Duplicate records and irrelevant information are removed to improve the quality of the dataset.

Handling Missing Values: Missing or incomplete data entries are identified and properly handled.

Text Processing: Customer reviews are processed to remove unnecessary words and convert text into a structured format.

Feature Extraction: Important features such as review timing, rating patterns, and reviewer activity are extracted for analysis.

Machine learning Model

Model Training: The system trains machine learning algorithms using historical review data to learn reviewer behavior patterns.

Reviewer Behavior Analysis: The model analyzes features such as review frequency, rating trends, and user activity levels.

Prediction of Early Reviewers: Based on learned patterns, the model predicts users who are likely to provide early reviews for newly launched products.

Natural Language Processing Module

Sentiment Analysis: The system analyzes review text to determine whether customer opinions are positive, negative, or neutral.

Keyword Extraction: Important keywords and product features mentioned in reviews are extracted.

Opinion Analysis: The system analyzes user opinions to understand customer preferences and feedback.

Personalized Recommendations and Risk Detection

Personalized Product Recommendations: The system recommends products to users based on their past review behavior and interests.

User Preference Analysis: User activity and previous reviews are analyzed to understand customer preferences.

Risk Detection: The system identifies suspicious or spam review activities that may affect review reliability.

Mental Health Support

User-Friendly Environment: The system provides a comfortable and supportive platform where users can freely share their opinions and experiences about products through reviews and ratings.

Encouraging Positive Feedback: Users are encouraged to provide constructive feedback, which helps create a positive and supportive online community.

Sentiment Understanding: By analyzing review sentiments, the system can understand user emotions and opinions related to products and services.

Advantages

Early Reviewer Identification: The system helps identify users who are likely to review products early.

Improved Marketing Strategy: Businesses can promote products more effectively during the early stage of product launch.

Better Review Analysis: Machine learning techniques improve the accuracy of analyzing reviewer behavior.

Spam Detection: The system helps detect fake or suspicious reviews.

3.3 SYSTEM REQUIREMENT SPECIFICATION

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 2 GB (min)
- Hard Disk: 40 GB
- Input Devices: Standard keyboard and mouse
- Display: SVGA monitor (min 21-inch)

Software Requirements

- Operating System: Windows 8 / 10 / 11
- Coding Language: Python
- Frontend: HTML, CSS, JavaScript
- Backend: Python
- Database: MySQL
- IDE / Tools: Visual Studio Code
- Web Browser: Google Chrome / Microsoft Edge / Firefox
- Framework: Django

3.4 SYSTEM STUDY

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are,

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

Legal and Ethical Feasibility

Compliance: The system follows rules and regulations related to data usage and uses publicly available review data only for research purposes.

Customer Privacy: The system protects user privacy by not collecting or exposing personal information.

Transparency: The system clearly explains how review data is used for predicting early reviewers.

Legal Risks: Possible risks include misuse of data or violation of platform policies, which are avoided by following proper data handling practices.

4. SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The System Architecture Diagram represents the overall structure of the proposed early reviewer prediction system. It shows how different components such as user interface, admin module, review database, preprocessing module, feature extraction module, machine learning prediction module, and result display module interact with each other. The architecture diagram provides a clear overview of the data flow and communication among the different parts of the system. It helps in understanding how the system processes review data and generates prediction results in an organized and efficient manner.

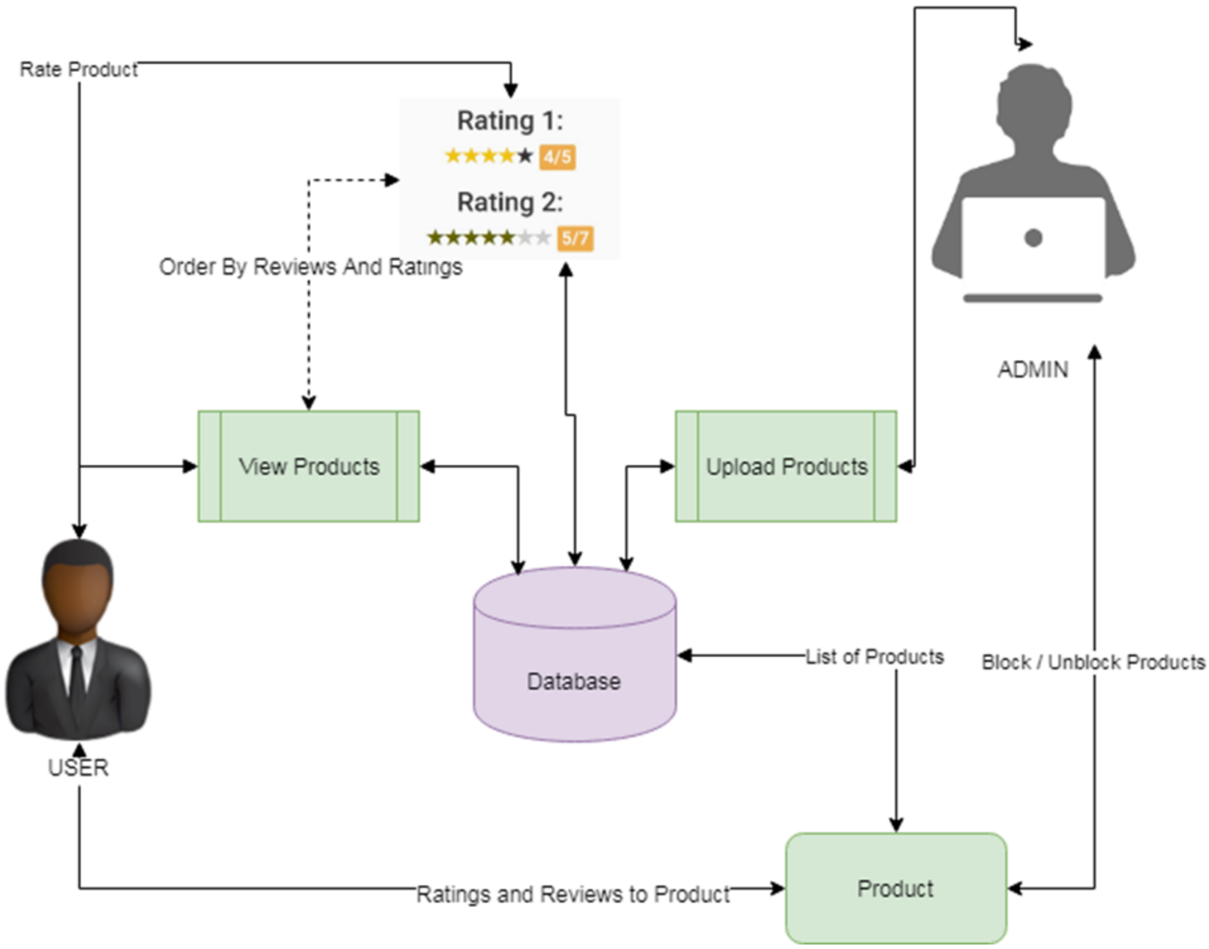


Fig 4.1 – System Architecture

4.2 UML DIAGRAMS

Unified Modeling Language (UML) is a standardized modeling language used to visualize, design, and document the components of a software system. It provides a clear graphical representation of how different parts of the system interact with each other. UML diagrams help developers, designers, and stakeholders understand the structure and behaviour of the system before actual implementation. By representing the system visually, it becomes easier to identify relationships between different modules, data flow, and user interactions within the system.

In this project, UML diagrams play an important role in explaining the working of the ecommerce platform designed to characterize and predict early reviewers. Diagrams such as Use Case Diagram, Class Diagram, Activity Diagram, and Sequence Diagram help illustrate how users interact with the system, how data is processed, and how different modules communicate with each other. These diagrams simplify complex system processes and provide a structured approach to system development. As a result, UML diagrams improve system planning, reduce development errors, and ensure that the project is implemented in an organized and efficient manner.

GOALS OF UML DIAGRAMS

The Primary goals in the design of the UML are as follows:

1. To clearly represent the structure and architecture of the system.
2. To describe how different modules and components interact with each other.
3. To simplify complex system processes through visual representation.
4. To help developers understand system requirements and functionality.
5. To provide a clear blueprint for designing and developing the system.
6. To improve communication between developers, designers, and users.
7. To identify possible design issues before system implementation.
8. To make the system documentation clearer and more understandable.

4.3 USE CASE DIAGRAM

A Use Case Diagram is used to represent the interaction between the actors and the system. It identifies the different functions provided by the system and shows how users interact with them. In this project, the main actors are the User and the Admin. The user can perform activities such as viewing products, checking ratings and reviews, and exploring product details. The admin is responsible for uploading product and review data, managing records, preprocessing data, applying machine learning algorithms, and viewing prediction results. The use case diagram gives a clear understanding of system functionality and user interaction.

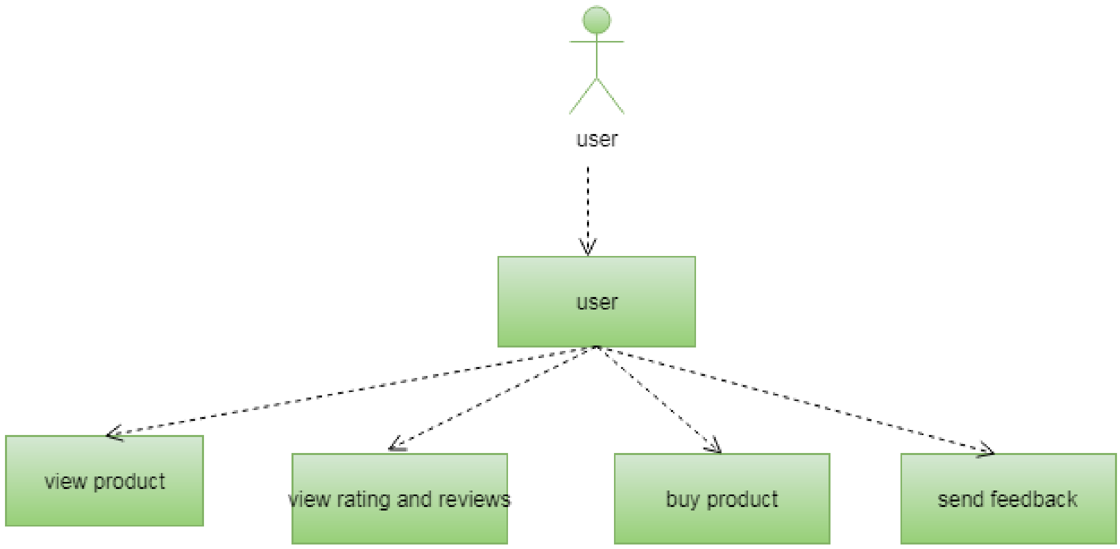


Fig 4.2 - Use Case Diagram for User

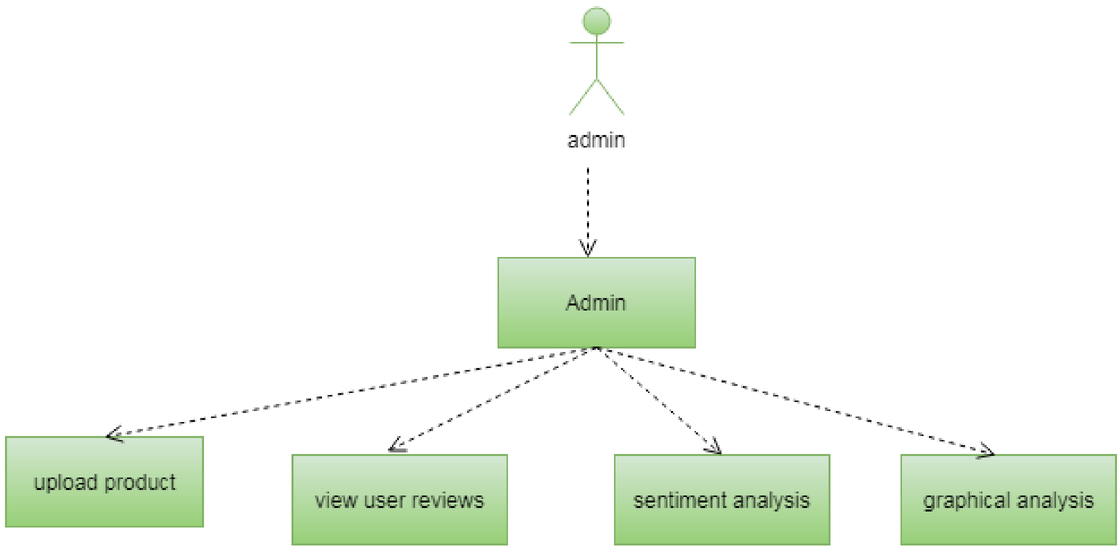


Fig 4.3 - Use Case Diagram for Admin

4.4 ACTIVITY DIAGRAM

An Activity Diagram is used to represent the workflow of the system by showing the sequence of activities performed by the actors and the system. It explains how the process flows from one step to another. In this project, the activity diagram describes actions such as user login, viewing products, checking ratings and reviews, admin login, uploading review data, preprocessing the dataset, extracting features, applying machine learning algorithms, and viewing prediction results. It helps in understanding the working flow of the proposed system step by step.

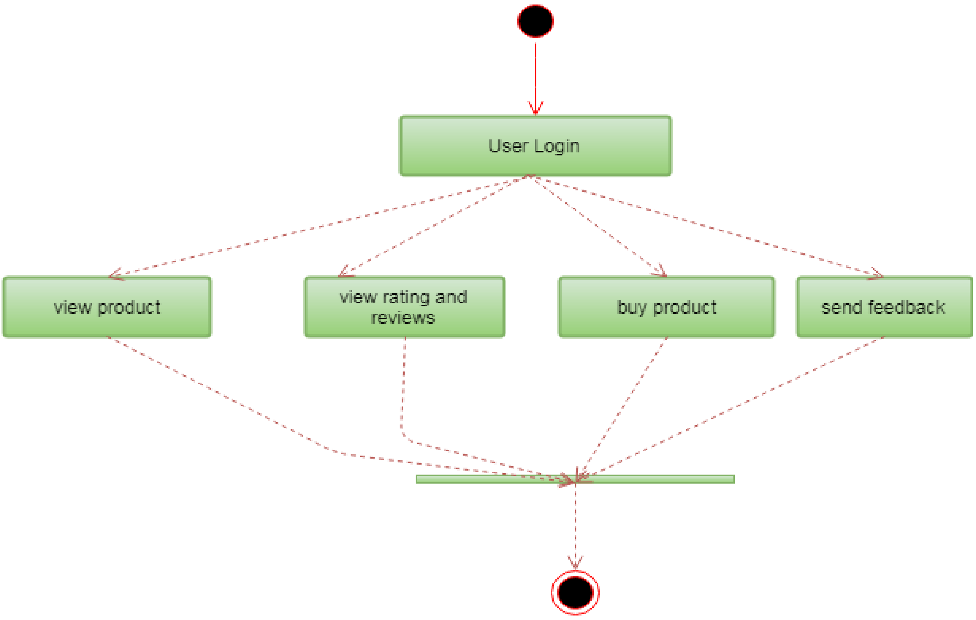


Fig 4.4 – Activity Diagram for User

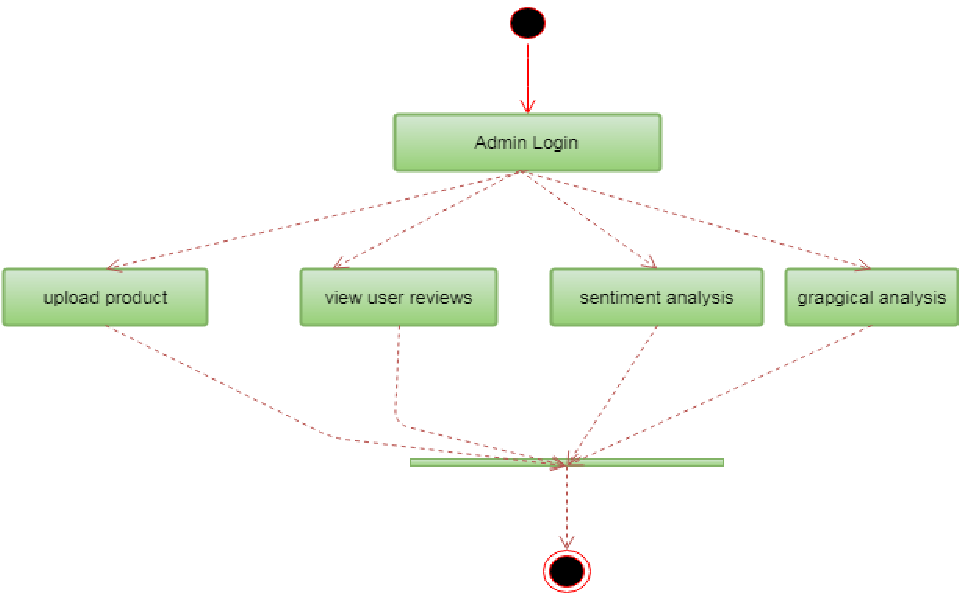


Fig 4.5 - Activity Diagram for Admin

4.5 SEQUENCE DIAGRAM

A Sequence Diagram is used to show the order in which operations take place in the system over time. It explains how different actors and system components communicate with each other to complete a task. In this project, the sequence diagram shows the interaction between the user, admin, system, and database. It includes actions such as login, viewing products, viewing ratings and reviews, uploading review data, preprocessing data, generating predictions, and displaying results. The sequence diagram helps in understanding the time-based interaction between the different parts of the system.

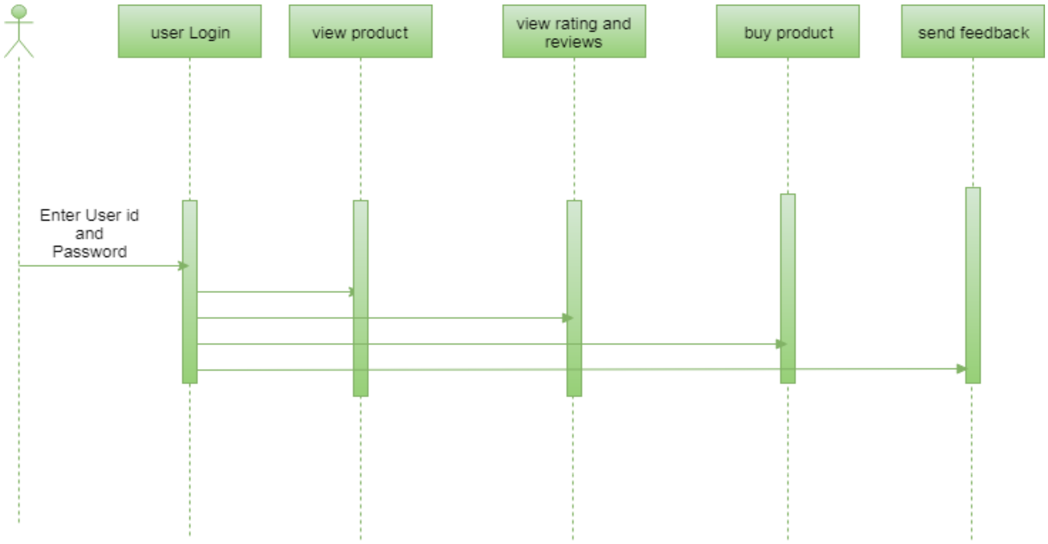


Fig 4.6 – Sequence Diagram for User

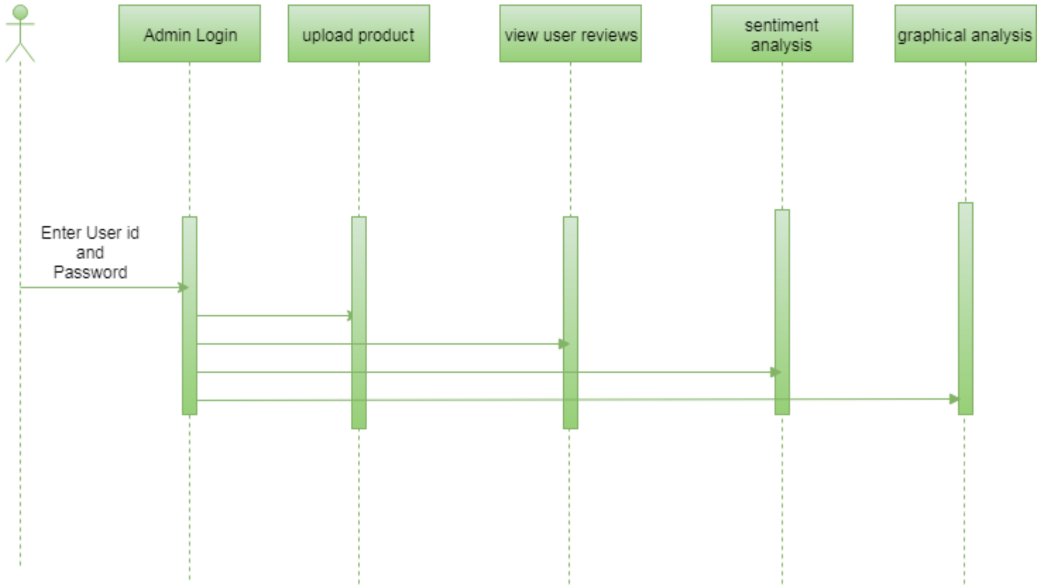


Fig 4.7 – Sequence Diagram for Admin

4.6 CLASS DIAGRAM

The Class Diagram represents the structure of the system by showing the main classes, their attributes, and their functions. In this project, the major classes are User, Admin, Product, Review, Dataset, and Prediction. The User class includes operations such as login, view products, and view reviews. The Admin class performs tasks such as managing products, managing review data, preprocessing the dataset, and viewing prediction results. The Product and Review classes store information related to products and customer reviews, while the Dataset and Prediction classes support data processing and machine learning operations. The class diagram helps in understanding the internal structure of the proposed system.

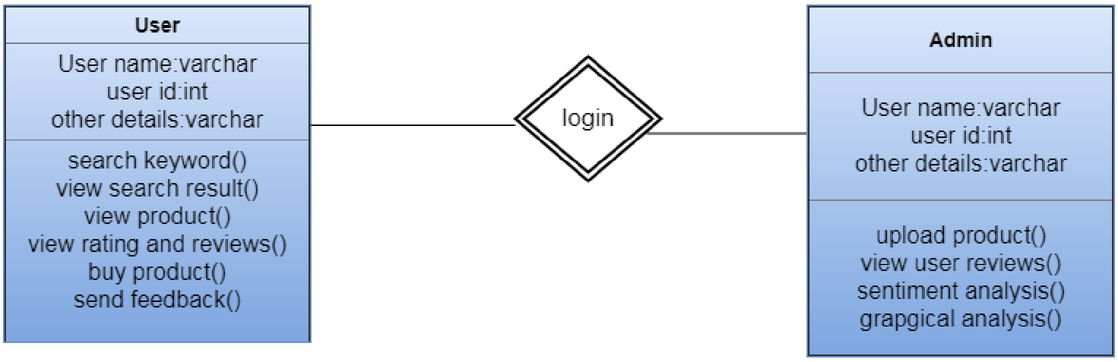


Fig 4.8 – Class Diagram

4.7 COMPONENT DIAGRAM

A Component Diagram is used to represent the physical and logical components of the system and the relationships between them. It shows how different parts of the application are organized and how they interact to perform the required operations. In this project, the component diagram explains the major system components such as user interface, admin panel, review management, data preprocessing, feature extraction, prediction module, database, and result analysis. It helps in understanding the modular structure of the system and provides a high-level view of the implementation.

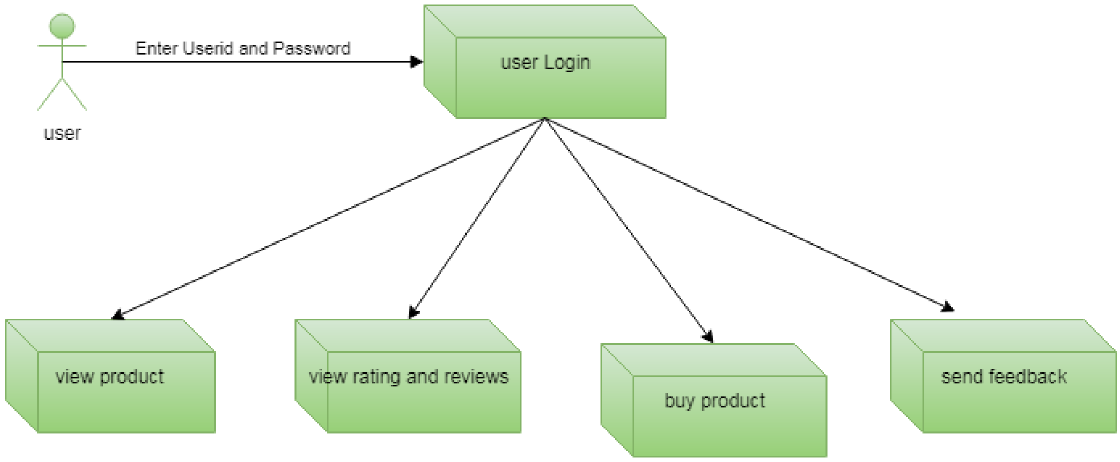


Fig 4.9 – Component Diagram for User

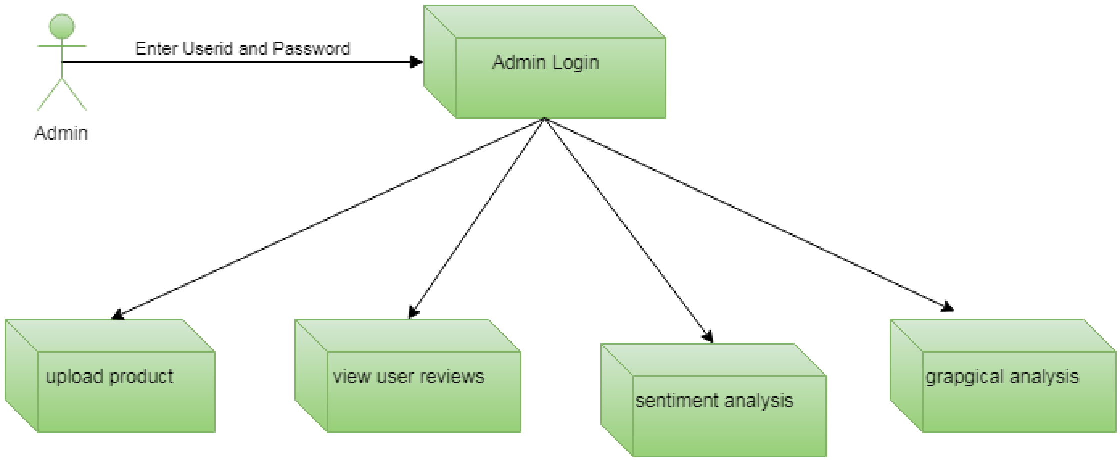


Fig 4.10 – Component Diagram for Admin

4.8 ER DIAGRAM

An Entity Relationship (ER) Diagram is used to represent the data structure of the system. It shows the entities involved in the project, their attributes, and the relationships between them. In this project, the ER diagram includes entities such as User, Admin, Product, Review, Dataset, and Prediction. It explains how user details, product information, review records, and prediction results are stored and connected in the database. The ER diagram helps in designing the database in a clear and structured manner.

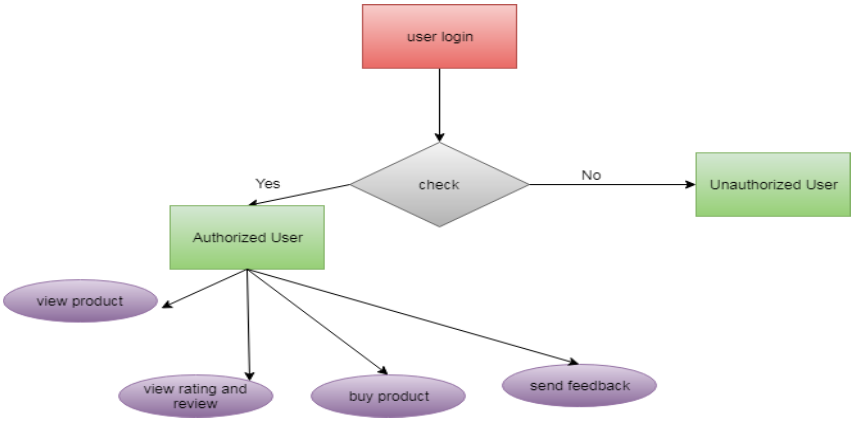


Fig 4.11 – ER Diagram for User

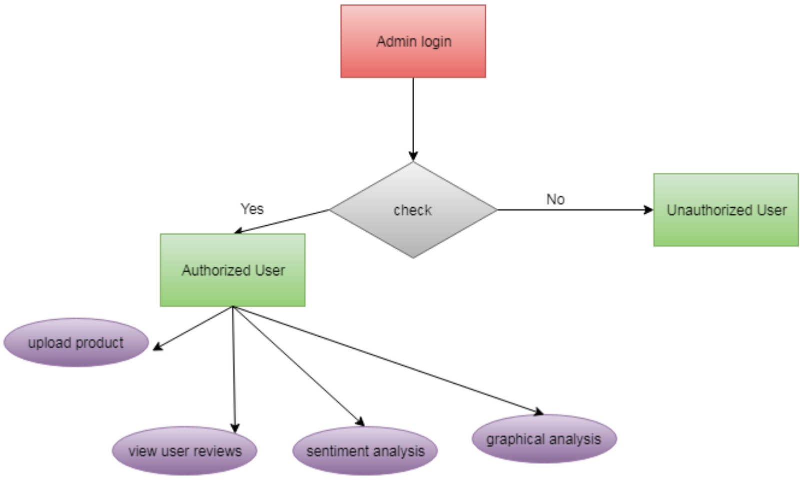


Fig 4.12 – ER Diagram for Admin

4.9 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is used to represent the flow of data within the system. It shows how data moves between users, processes, and the database. In this project, the DFD explains how review data is collected, stored, processed, and used for prediction. It also shows how the admin interacts with the system to upload data, preprocess it, extract features, and generate early reviewer predictions. The DFD helps in understanding the movement of information clearly and systematically.

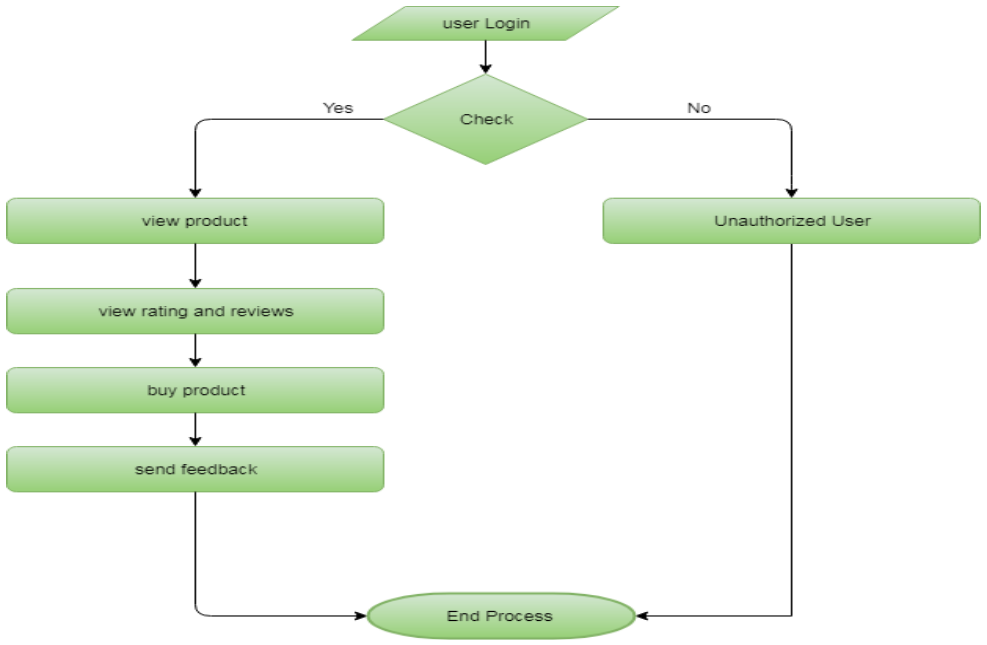


Fig 4.13 – Data Flow Diagram for User

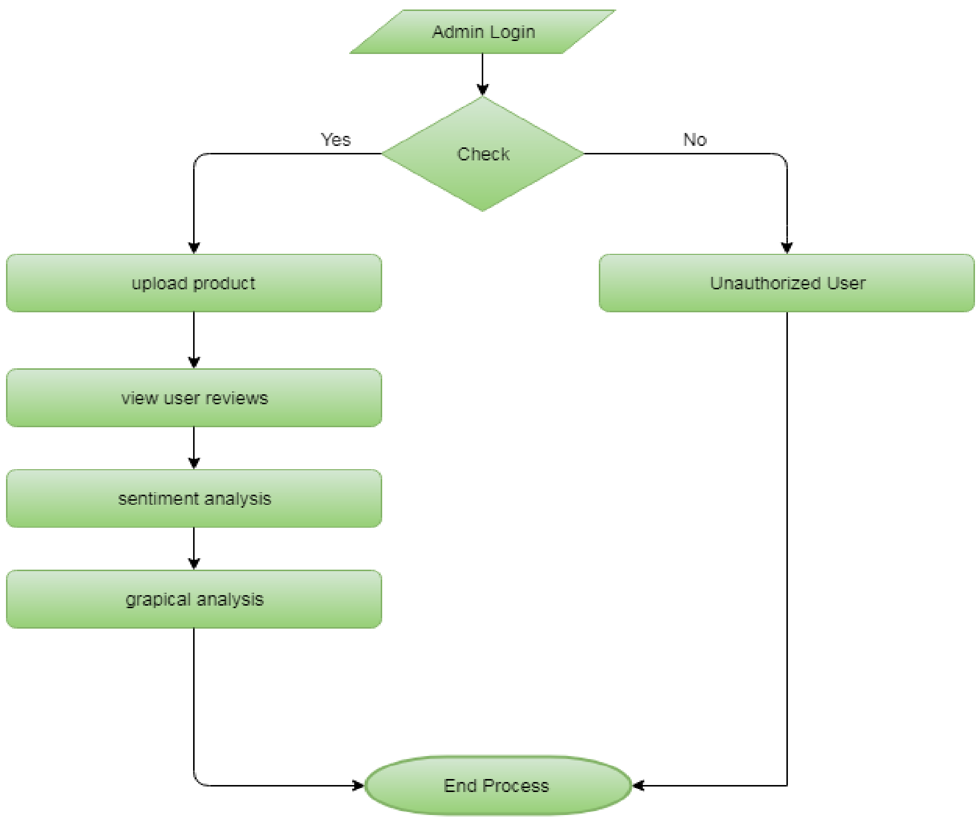


Fig 4.14 – Data Flow Diagram for Admin

5. IMPLEMENTATION

5.1 ABOUT PYTHON

Python is a high-level, interpreted, and general-purpose programming language widely used in modern software development. It is one of the most popular programming languages because of its simple syntax, readability, and ease of use. Python supports multiple programming paradigms such as procedural programming, object-oriented programming, and functional programming. Due to these features, Python is used in many areas including web development, data analysis, artificial intelligence, machine learning, automation, and scientific computing.

One of the major advantages of Python is its large collection of standard libraries and third-party packages. These libraries reduce coding complexity and help developers build applications quickly and efficiently. Python is also platform-independent, which means programs written in Python can run on different operating systems such as Windows, Linux, and macOS with very few changes. Its easy syntax allows developers to write fewer lines of code compared to many other programming languages, which improves productivity and simplifies maintenance.

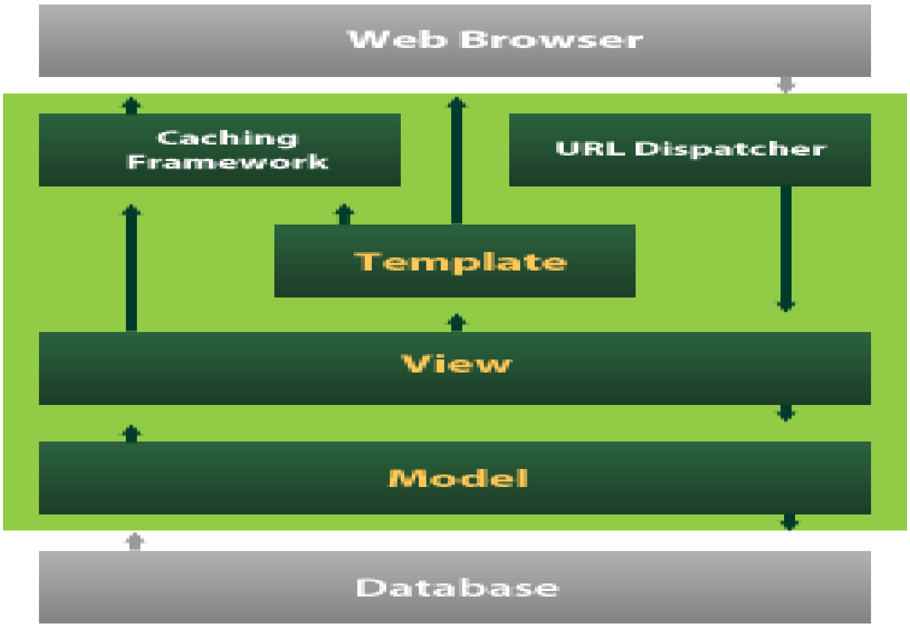
In this project, Python is used as the core programming language for developing the early reviewer prediction system. It plays an important role in handling the complete workflow of the project, including data preprocessing, feature extraction, classification, prediction, and integration with the web application. Python is highly suitable for this project because it offers strong support for machine learning and data processing tasks.

Python libraries such as NumPy and Pandas are used to store, process, and manipulate large amounts of review data efficiently. Scikit-learn is used for implementing machine learning algorithms such as Logistic Regression, Random Forest, and Support Vector Machine. These libraries make it easier to build predictive models and evaluate their performance using standard metrics. Python also supports database connectivity and web development frameworks, which helps in creating a complete and user-friendly application.

Another important reason for selecting Python is its flexibility. It allows easy integration with front-end technologies, databases, and machine learning modules. This makes Python an ideal choice for developing intelligent e-commerce applications. Therefore, Python provides the required efficiency, simplicity, and scalability for implementing the proposed early reviewer prediction system successfully.

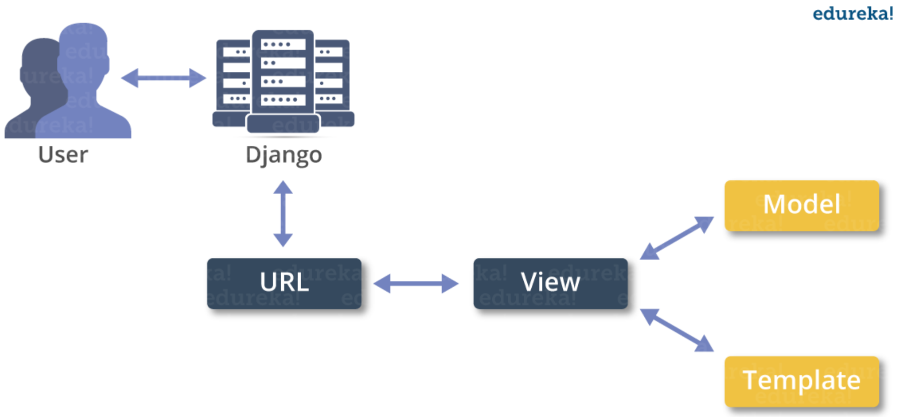
5.2 ABOUT DJANGO

Django is a high-level Python web framework that encourages rapid development and clean, pragmatic design. It is free and open-source and is mainly used for building secure and database-driven web applications. Django reduces the complexity of web development by providing built-in features such as URL routing, database handling, authentication, form processing, and admin panel support. It follows the principle of reusability and helps developers create applications in an organized and efficient manner.



In this project, Django is used as the web framework for developing the application interface and connecting the front end with the back end. It helps in handling user requests, database operations, admin functions, and the display of prediction results through the web interface. Django also provides an administrative interface that makes it easier to manage reviewer details, product information, review records, and prediction outputs.

Another important advantage of Django is that it supports rapid development and improves application security through built-in features. It also works efficiently with Python libraries and databases, which makes it suitable for machine learning-based web applications. Therefore, Django plays an important role in building the proposed early reviewer prediction system in an efficient, secure, and organized manner.



5.3 ABOUT MACHINE LEARNING

Machine Learning is a branch of artificial intelligence that allows computers to learn patterns from data and make decisions or predictions without being explicitly programmed for every individual task. Instead of following only fixed instructions, machine learning systems use historical data to identify patterns and build models that can be applied to new data. This makes machine learning highly useful in solving real-world problems that involve prediction, classification, recommendation, and pattern recognition.

Machine learning has become very important in many application areas such as healthcare, banking, marketing, security, e-commerce, and education. In e-commerce platforms, machine learning is used for recommendation systems, fraud detection, customer segmentation, sentiment analysis, product ranking, and review classification. It helps organizations make data-driven decisions and automate complex processes efficiently.

In this project, machine learning is used to predict early reviewers in e-commerce platforms. The main goal is to identify users who are likely to post reviews during the early stage of a product launch. This prediction is made by analyzing historical review records and extracting meaningful features related to user behaviour and review timing.

The model learns from past data and then predicts whether a particular user is likely to become an early reviewer.

The machine learning process in this project includes several steps such as data collection, data preprocessing, feature extraction, model training, testing, and evaluation. After the dataset is prepared, relevant features like review frequency, rating behaviour, review count, activity level, and review time gap are selected. These features are given to machine learning algorithms, which learn the differences between early reviewers and regular reviewers.

Supervised learning algorithms are mainly used in this project because the available review data can be labelled into early reviewers and non-early reviewers. Algorithms such as Logistic Regression, Random Forest, and Support Vector Machine are suitable for classification tasks. Once trained, these models can predict reviewer categories for new data. The results are then evaluated using metrics such as accuracy, precision, recall, and F1-score.

Machine learning improves the system by identifying hidden patterns that may not be visible through manual analysis. It reduces human effort, increases prediction accuracy, and helps businesses take proactive steps during product launch stages. Therefore, machine learning serves as the intelligence behind the proposed system and makes it capable of supporting effective product marketing strategies in e-commerce platforms.

5.4 MODULES

The Characterization and Prediction of Early Reviewers for Product Marketing in E-Commerce Platforms system is divided into several modules. Each module performs a specific task in managing products, users, orders, reviews, and analysis. The system allows users to browse products, place orders, submit ratings and reviews, and helps the administrator analyze customer opinions through charts and sentiment analysis. The main modules of the system are described below.

Module 1: User Interface Module

This module provides an interface for users and administrators to interact with the system. It allows users to register, login, browse products, add items to cart, place orders, and submit ratings and reviews. It also allows the administrator to login and manage products and analysis.

Functions

- Provide login and registration pages
- Display product details to users
- Allow users to add products to cart
- Allow users to place orders
- Allow users to submit ratings and reviews
- Provide admin interface for product management and chart analysis

Module 2: User Management Module

This module is responsible for handling user registration and login operations. New users can create an account by entering their details, and existing users can access the system using valid credentials. This module helps in maintaining user data and session handling.

Functions

- Register new users
- Validate user login credentials
- Maintain user session details
- Store user profile information
- Allow secure access to user pages

Module 3: Product Management Module

This module is used to store and manage product-related details in the system. The administrator can upload new products and delete unwanted products. Users can view the uploaded products along with their complete information such as name, vendor, version, color, price, features, and images.

Functions

- Upload new product details
- Store product name, vendor, version, and color

- Store product price and features
- Display product images
- Delete products when required
- Show products to users

Module 4: Cart and Order Management Module

This module manages the shopping cart and purchase process. Users can add products to the cart, select quantity, enter delivery details, and place orders. The system also stores order history and order details for future reference.

Functions

- Add products to cart
- Manage product quantity
- Calculate total price
- Accept delivery details
- Complete checkout and purchase process
- Store order history
- Display order details to users

Module 5: Review and Rating Module

This module enables users to submit product ratings and textual reviews. It stores feedback given by users and displays the reviews for other users to view. This module plays an important role in collecting customer opinions about products.

Functions

- Submit star ratings
- Submit review comments
- Store user feedback in database
- Display ratings and reviews
- Support product opinion analysis

Module 6: Sentiment Analysis Module

This module is responsible for analyzing the review text submitted by users. The system compares the review comment with predefined positive and negative words and classifies the sentiment as positive, negative, or neutral. This helps in understanding customer satisfaction towards a product.

Functions

- Read review text
- Compare review words with positive word list
- Compare review words with negative word list
- Classify sentiment as positive, negative, or neutral
- Store sentiment result for analysis

Module 7: Chart Analysis Module

This module provides graphical analysis of user reviews and ratings. The administrator can view charts based on product-wise review count, profession-wise opinions, location-wise opinions, and sentiment-wise analysis. This helps in understanding user behaviour and product performance more effectively.

Functions

- Generate product-wise review charts
- Generate profession-wise opinion charts
- Generate region-wise opinion charts
- Generate sentiment-wise charts
- Display graphical analysis to admin

5.5 SYSTEM IMPLEMENTATION PROCESS

The implementation of the proposed system is carried out in a sequence of well-defined steps. First, the historical review dataset is collected and stored in the database. This dataset contains reviewer details, product information, ratings, review text, timestamps, and other required attributes.

Next, the dataset is preprocessed to remove missing values, duplicate entries, and inconsistencies. Data cleaning and transformation are performed so that the dataset becomes suitable for machine learning analysis. After preprocessing, important behavioural and temporal features are extracted from the cleaned dataset.

The processed data is then divided into training and testing sets. Machine learning algorithms such as Logistic Regression, Random Forest, and Support Vector Machine are trained using the training data. These models learn patterns that distinguish early reviewers from regular reviewers.

After training, the models are tested using unseen data to evaluate their performance. The results are measured with the help of accuracy, precision, recall, and F1-score. Based on the evaluation, the best-performing model is selected for reviewer prediction.

Finally, the trained model is integrated into the application. The system displays the prediction results through the user interface, which allows the administrator to identify likely early reviewers and use this information for better marketing and product promotion strategies.

5.6 LIBRARIES AND TOOLS USED

The implementation of the proposed system uses several libraries and tools that support data analysis, machine learning, and web development.

Python 3.7.11:
Used as the main programming language for developing the complete system.
Django:
Used as the web framework for developing the application and connecting the front end with the back end.
NumPy:
Used for numerical computations and array operations.
Pandas:
Used for data loading, preprocessing, and manipulation.
Scikit-learn:
Used for implementing machine learning algorithms and performance evaluation.
MySQL:
Used for storing user details, product information, and review records.
HTML, CSS, JavaScript:
Used for developing the front-end interface of the application.

VS Code:

Used as the development environment for coding and testing the project.

5.7 CODING

File Name: Views.py

The views.py file contains the main logic of the web application. It handles user requests, processes product-related operations, manages user sessions, and returns responses to the web pages. This file acts as a bridge between the front end, database, and application logic.

The file performs functions such as:

- User login validation
- User registration
- Product display
- Product purchase and cart management
- Ratings and reviews submission
- Sentiment analysis of reviews
- Order details display
- Logout handling

Code:

```
from django.shortcuts import render, redirect, get_object_or_404


from admins.models import Prodcuts
from user.forms import UsersForm, PurchaseForm
from user.models import Users, Purchase, Feedback
from user.words import positive_words, negative_words
from django.contrib import messages


def home(request):
    if 'admin' in request.session:
        del request.session['admin']
```

```
    products = Products.objects.all()
    return render(request, 'user/home.html', {'products': products})

def viewproduct(request, pk):
    pro = get_object_or_404(Prodcuts, id=pk)

    if request.method == "POST":

        if 'userid' not in request.session:
            messages.warning(request, "Please login first to add items to cart.")
            return redirect('user:index')

        uid = request.session['userid']
        uses = get_object_or_404(Users, id=uid)
        form = PurchaseForm(request.POST)

        if form.is_valid():
            ff = form.save(commit=False)
            ff.customer = uses
            ff.purhased = pro
            ff.totalprice = pro.price * ff.quantity
            ff.save()
            messages.success(request, "Item added to cart successfully!")
            return redirect('user:cart')
    else:
        form = PurchaseForm()

    return render(request, 'user/viewproduct.html', {'prod': pro, 'ipk': pk, 'form': form})

def cart(request):
    if 'userid' not in request.session:
        messages.info(request, "Please login to view your cart and checkout.")
        return redirect('user:index')
```

```
    uid = request.session['userid']
    uses = get_object_or_404(Users, id=uid)
    p = Purchase.objects.filter(customer=uses, status='incart')
        orders = Purchase.objects.filter(customer=uses, status='purchased').order_by('-
created_at')

    if request.method == "POST":
        if 'userid' not in request.session:
            messages.error(request, "Session expired. Please login again.")
            return redirect('user:index')

        action = request.POST.get('action', '')
        delivery_address = request.POST.get('delivery_address', '')
        delivery_mobile = request.POST.get('delivery_mobile', '')

        if action == 'buynow':
            if not delivery_address or not delivery_mobile:
                messages.error(request, "Please enter delivery address and mobile number.")
                return render(request, 'user/cart.html', {'p': p, 'orders': orders})

            Purchase.objects.filter(customer=uses, status='incart').update(
                status='purchased',
                delivery_address=delivery_address,
                delivery_mobile=delivery_mobile
            )
            messages.success(request, "Payment successful! Your order has been placed.")
            p = Purchase.objects.filter(customer=uses, status='incart')
             orders = Purchase.objects.filter(customer=uses, status='purchased').order_by('-
created_at')
            return render(request, 'user/cart.html', {'p': p, 'orders': orders})
        else:
            Purchase.objects.filter(customer=uses, status='incart').update(status='checkout')
            return redirect('user:home')
```

```
    return render(request, 'user/cart.html', {'p': p, 'orders': orders})

def viewratings(request, pk):
    pro = get_object_or_404(Prodcuts, pk=pk)
    fedbck = Feedback.objects.filter(product=pro)
    return render(request, 'user/viewratings.html', {'feedbacks': fedbck})

def addratings(request, pk):
    pos, neg = 0, 0
    sen = 'pending'
    pro = get_object_or_404(Prodcuts, pk=pk)
    uses = None
    stat = 'pending'

    try:
        uid = request.session['userid']
        uses = get_object_or_404(Users, id=uid)
    except:
        uses = None

    if uses:
        try:
            Purchase.objects.get(customer=uses, purhased=pro, status='purchased')
            stat = 'purchased'
        except:
            stat = 'not purchased'

    if request.method == "POST":
        if 'userid' not in request.session:
            messages.warning(request, "Please login first to add rating.")
            return redirect('user:index')

        ratings = request.POST.get('rating', '')
        comments = request.POST.get('comment', '')
```

```
        for pword in positive_words:
            if pword in comments:
                pos += 1

        for nword in negative_words:
            if nword in comments:
                neg += 1

        if pos > neg:
            sen = 'positive'
        elif neg > pos:
            sen = 'negative'
        else:
            sen = 'neutral'

        Feedback.objects.create(
            user=uses,
            product=pro,
            isPurchased=stat,
            rating=ratings,
            review=comments,
            sentiment=sen
        )
        messages.success(request, "Rating submitted successfully!")
        return redirect('user:home')

    return render(request, 'user/addratings.html', {'pro': pro})

def index(request):
    message = None
    if request.method == "POST":
        username = request.POST.get('username').strip()
        password = request.POST.get('password').strip()
```

```
        users = Users.objects.filter(username=username, password=password).first()

        if users:
            request.session.flush()
            request.session['userid'] = users.id
            request.session['username'] = users.username
            return redirect('user:home')
        else:
            messages.error(request, "User name and password are not matching...")
            message = "User name and password are not matching..."

    return render(request, 'user/index.html', {'msg': message})

def registration(request):
    if request.method == "POST":
        loca = request.POST.get('location', '')
        users = UsersForm(request.POST)

        if users.is_valid():
            formss = users.save(commit=False)
            formss.location = loca
            formss.save()
            messages.success(request, "Registered successfully!")
            return redirect('user:index')
    else:
        users = UsersForm()

    return render(request, 'user/registration.html', {'form': users})

def order_detail(request, pk):
    if 'userid' not in request.session:
        messages.info(request, "Please login to view order details.")
        return redirect('user:index')
```

```
    uid = request.session['userid']
    uses = get_object_or_404(Users, id=uid)
    order = get_object_or_404(Purchase, id=pk, customer=uses)
    return render(request, 'user/order_detail.html', {'order': order})

def logout(request):
    request.session.flush()
    messages.success(request, "You have been logged out successfully!")
    return redirect('user:index')
```

Code Explanation

1. Importing Required Libraries

The program begins by importing the required Django functions, models, forms, and sentiment word lists.

- render, redirect, and get_object_or_404 are used for page rendering and redirection.
- Prodcuts model is used to access product details.
- UsersForm and PurchaseForm are used for registration and purchase forms.
- Users, Purchase, and Feedback models are used to store user, order, and review details.
- positive_words and negative_words are used for sentiment analysis.
- messages is used to display notifications.

2. Home Page Function

The home() function displays all products available in the system.

- Retrieves product details from the database.
- Sends product data to the home page.
- Helps users browse products.

3. View Product Function

The viewproduct() function displays a selected product and allows the user to add it to the cart.

- Retrieves product details using product ID.
- Checks whether the user is logged in.
- Accepts quantity through purchase form.
- Calculates total price.
- Saves purchase details in the database.

4. Cart Management Function

The cart() function manages cart items and purchased orders.

- Checks whether the user is logged in.
- Displays cart items and order history.
- Accepts delivery address and mobile number.
- Updates order status after checkout.

5. View Ratings Function

The viewratings() function displays ratings and feedback of a selected product.

- Retrieves the selected product.
- Fetches all feedback related to that product.
- Displays reviews on the ratings page.

6. Add Ratings and Review Function

The addratings() function allows users to submit ratings and reviews.

- Retrieves the selected product.
- Checks whether the user is logged in.
- Verifies whether the product was purchased.
- Accepts rating and review comment.
- Performs sentiment analysis.
- Stores feedback in the database.

The sentiment is classified as:

- **Positive** – if positive words are more
- **Negative** – if negative words are more
- **Neutral** – if both are equal

7. User Login Function

The index() function is used for user login.

- Accepts username and password.
- Checks credentials from the database.
- Stores user session details.
- Redirects the user to the home page.

8. User Registration Function

The registration() function creates a new user account.

- Accepts registration form data.
- Validates the form.
- Saves user details in the database.
- Redirects the user to the login page.

9. Order Detail Function

The order_detail() function displays full details of a particular order.

- Checks whether the user is logged in.
- Retrieves the selected order.
- Ensures the order belongs to the logged-in user.
- Displays order information.

10. Logout Function

The logout() function is used to end the user session.

- Clears session data.
- Logs the user out.
- Displays logout success message.
- Redirects to the login page.

6. SYSTEM TESTING

System testing is an important phase in software development. The main purpose of testing is to identify errors and ensure that the developed system works correctly according to the specified requirements. Testing helps verify whether the software system satisfies user expectations and performs its intended functions without failure. It is carried out at different levels to validate individual modules, integrated components, and the complete system.

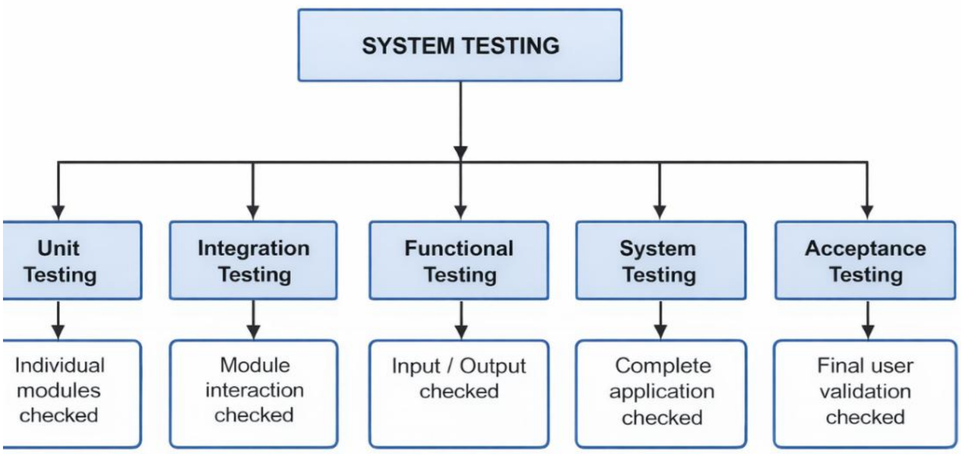


Figure: Types of Testing Used in the Proposed System

In this project, testing is performed to ensure that the early reviewer prediction system works properly in terms of user interaction, data processing, prediction accuracy, and overall system performance. Different types of testing are applied to check the functionality, integration, reliability, and usability of the system.

6.1 TYPES OF TESTS

6.1.1 Unit Testing

Unit testing is used to test the individual modules of the application separately. Each function or component is checked to ensure that it performs correctly according to the expected input and output. In this project, unit testing is carried out for modules such as user login, registration, review data handling, feature extraction, prediction process, and result display.

6.1.2 Integration Testing

Integration testing is performed to verify whether different modules of the system work together properly. It helps identify interface errors between components. In this project, integration testing checks the connection between the Django application, machine learning model, database, and user interface to ensure smooth data flow and proper interaction among modules.

6.1.3 Functional Testing

Functional testing is carried out to verify whether the system performs all intended functions according to the project requirements. It ensures that valid inputs are accepted, invalid inputs are handled properly, and the expected outputs are generated. In this project, functional testing includes checking user registration, login validation, data preprocessing, prediction generation, and dashboard display.

6.1.4 System Testing

System testing is performed on the complete integrated application to verify that the entire system works as expected. It ensures that all modules operate together correctly in a real execution environment. In this project, system testing is used to confirm that the complete early reviewer prediction system functions properly from data input to final prediction output.

6.1.5 White Box Testing

White box testing is a testing method in which the internal structure and logic of the code are examined. It is useful for checking internal program flow, conditions, loops, and decision-making statements. In this project, white box testing is applied while testing the internal logic of data preprocessing, feature extraction, and prediction functions.

6.1.6 Black Box Testing

Black box testing is a testing method in which the system is tested without considering the internal code structure. The focus is mainly on input and output behaviour. In this project, black box testing is used to verify user-facing functionalities such as login, registration, chatbot interaction, review analysis, and prediction result display.

6.1.7 Acceptance Testing

Acceptance testing is the final stage of testing performed to ensure that the system satisfies user requirements and is ready for deployment or demonstration. In this project, acceptance testing confirms that the developed system meets the expected objectives and produces useful prediction results for identifying early reviewers.

6.2 TEST STRATEGY AND APPROACH

The testing process for this project is mainly carried out manually. Each module is tested with different sets of inputs to verify its functionality and performance. Special attention is given to user inputs, database connectivity, machine learning prediction flow, and result generation. The testing approach ensures that the system is reliable, accurate, and easy to use.

6.3 OBJECTIVES OF TESTING

The main objectives of testing in this project are as follows:

- To verify that all modules work correctly.
- To ensure that valid user inputs are accepted properly.
- To confirm that invalid inputs are rejected with appropriate messages.
- To check whether all links and pages navigate correctly.
- To ensure that prediction results are generated and displayed properly.
- To verify that database operations such as insert, update, and retrieval work correctly.
- To ensure that the system performs efficiently without errors or delays.

6.4 FEATURES TO BE TESTED

The following features are tested in the proposed system:

- User registration and login
- Admin login and management functions
- Review data collection and storage
- Data preprocessing operations
- Feature extraction process
- Machine learning prediction module
- Result display and dashboard generation

- Navigation between web pages
- Database connectivity
- Input validation and duplicate entry checking

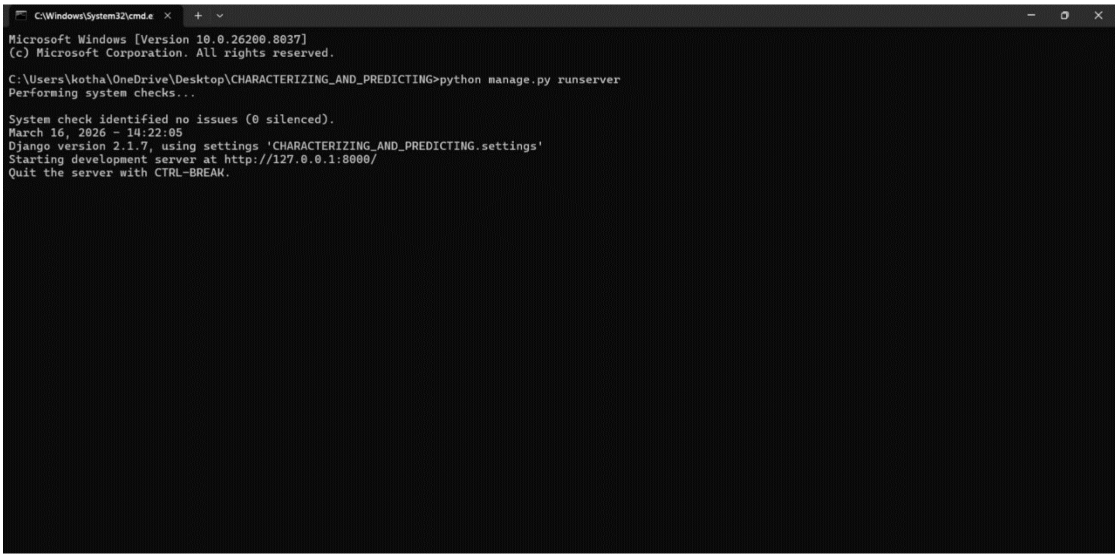
6.5 TEST RESULTS

The system was tested using different test cases for individual modules as well as the complete application. All major functions such as registration, login, data handling, prediction, and result display worked successfully. The integration between the Django application, machine learning model, and database was also tested successfully.

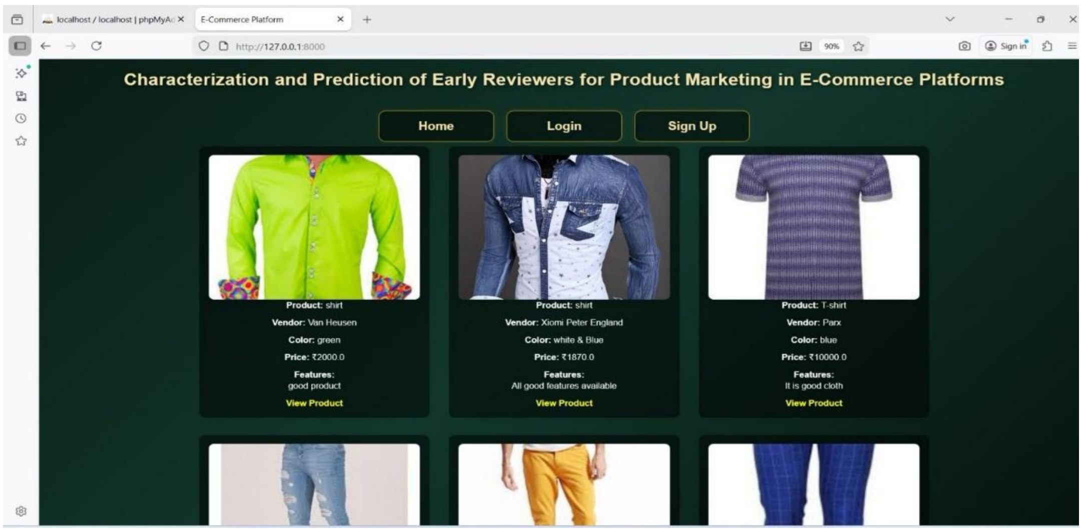
Result: All the test cases were executed successfully, and the system produced the expected output. No major defects were encountered during testing.

7. SCREENSHOTS

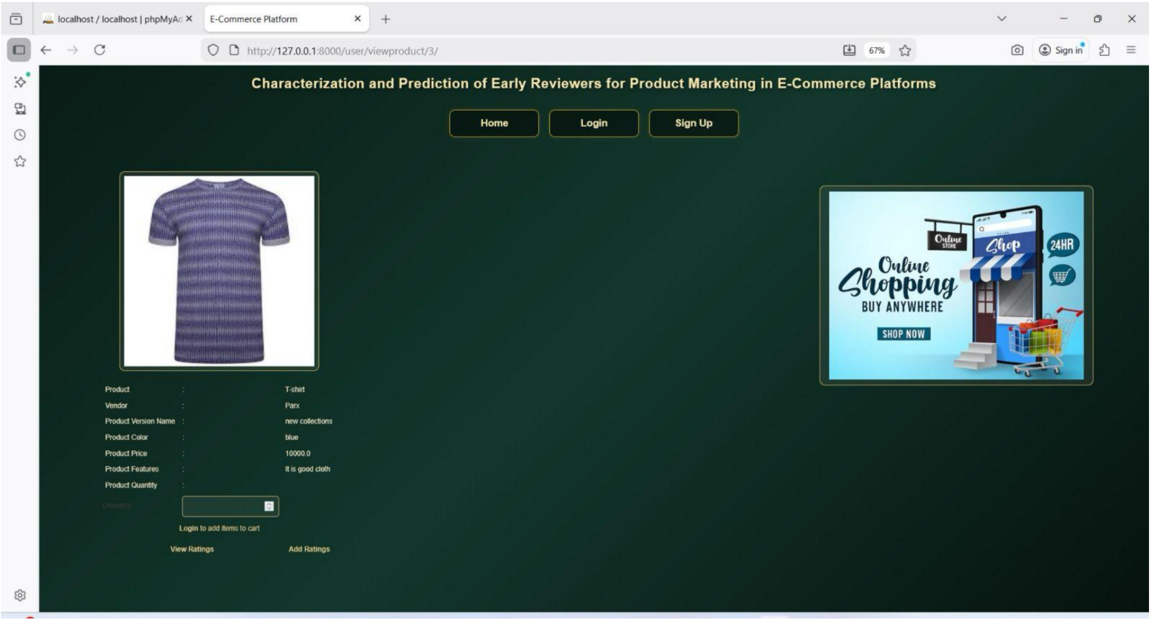
Step 1: Start WAMP Server and import the database using **phpMyAdmin**. Then open Command Prompt, navigate to the project folder, and run “python manage.py runserver”.



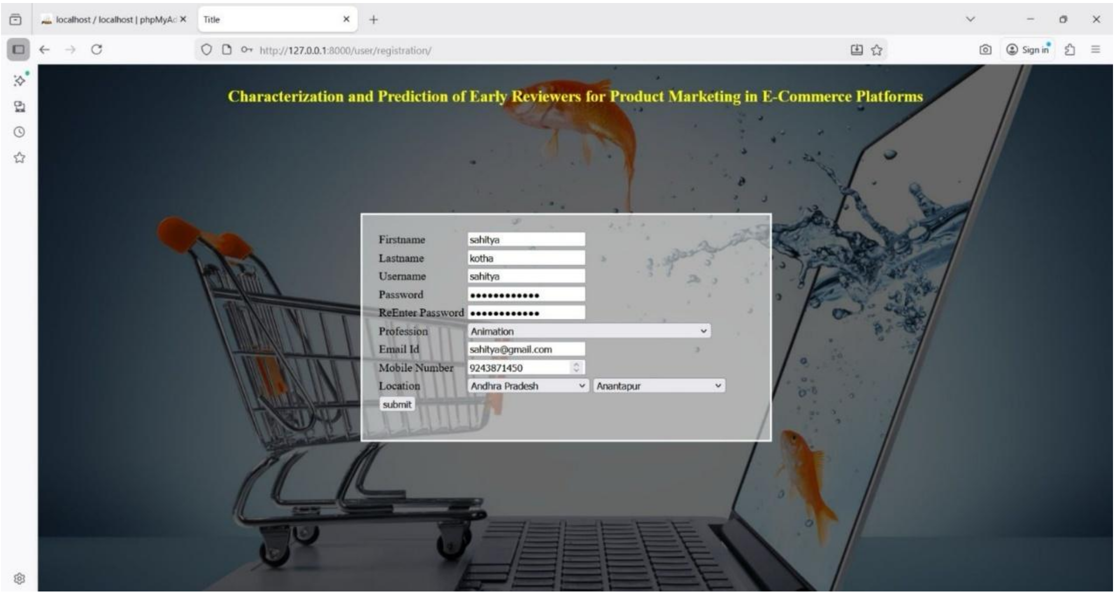
Step 2: After starting the server, open http://127.0.0.1:8000/ to view the home page. It displays available products with details and provides options for login and registration.



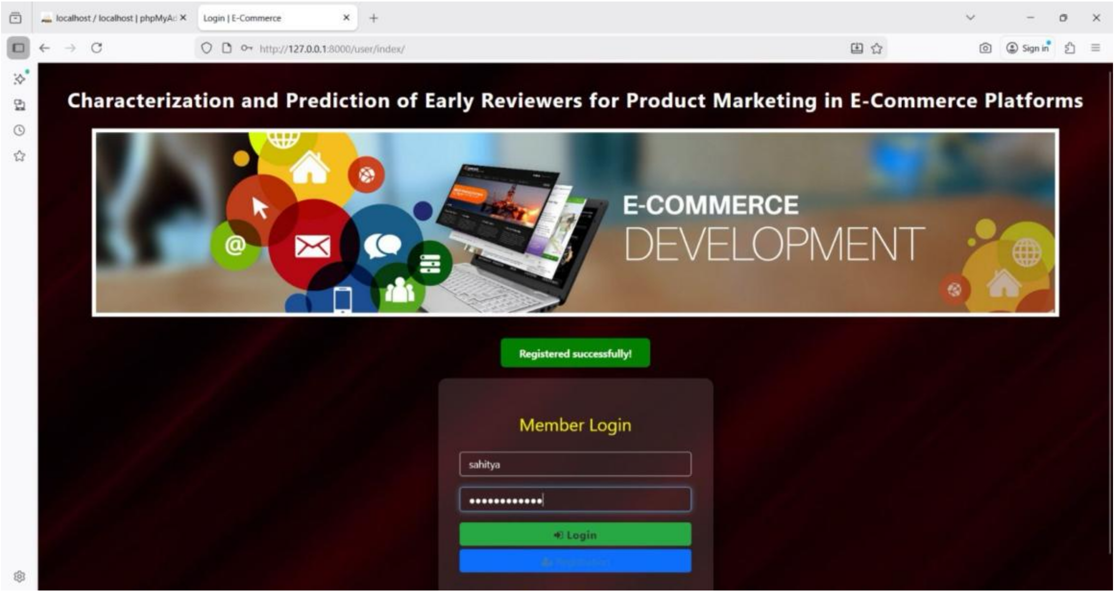
Step 3: This page displays detailed information about the selected product and allows users to select quantity, add to cart, and view or add ratings.



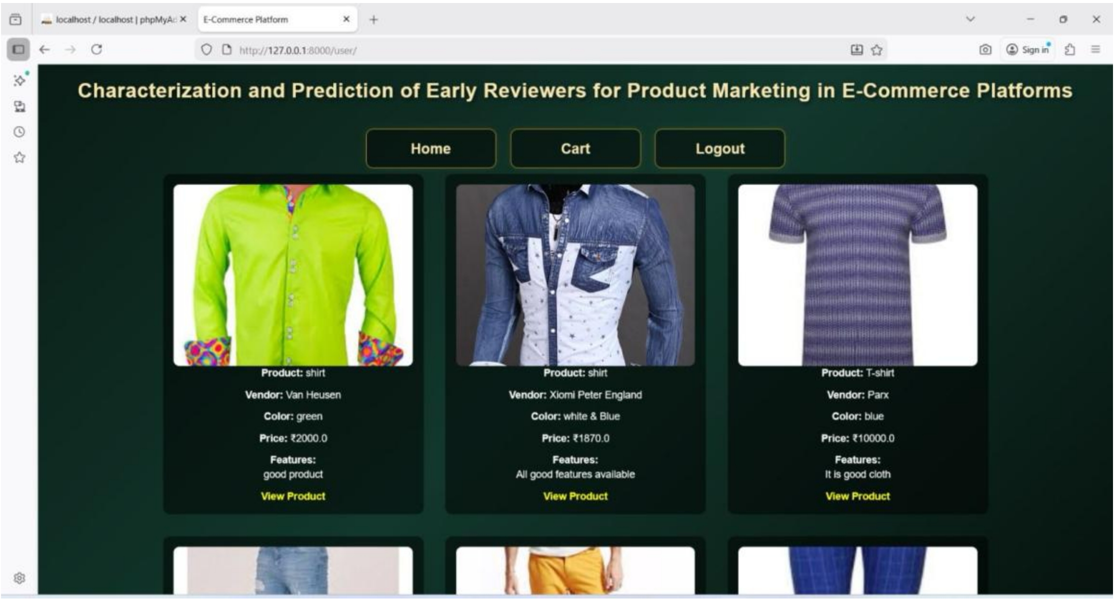
Step 4: This page allows new users to enter their personal details and complete the registration process to create an account.



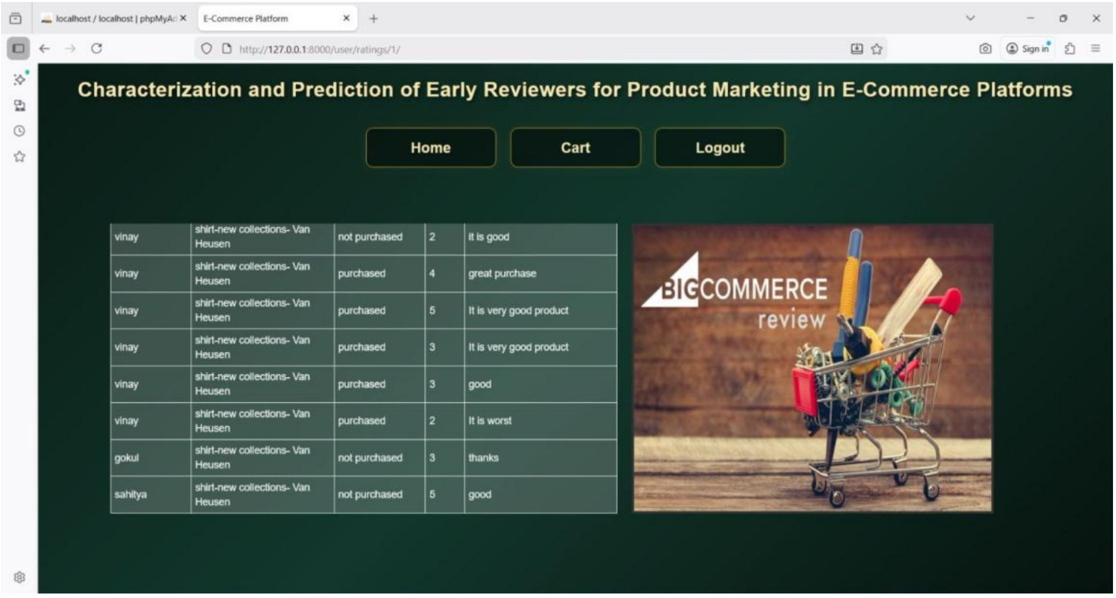
Step 5: This page allows users to enter their login credentials and access the system successfully.



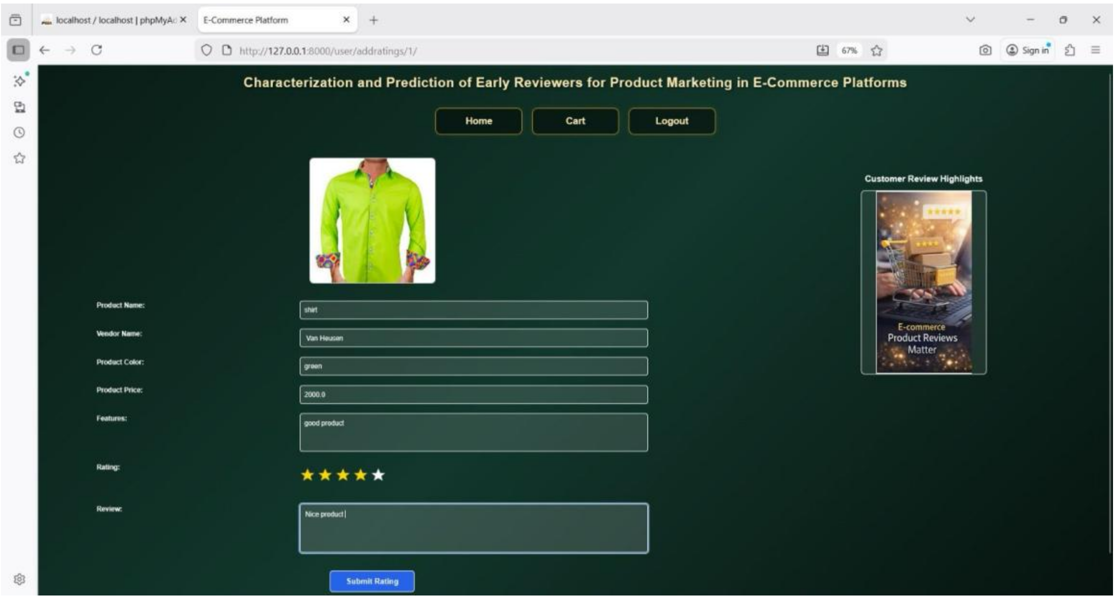
Step 6: This page displays available products after user login.



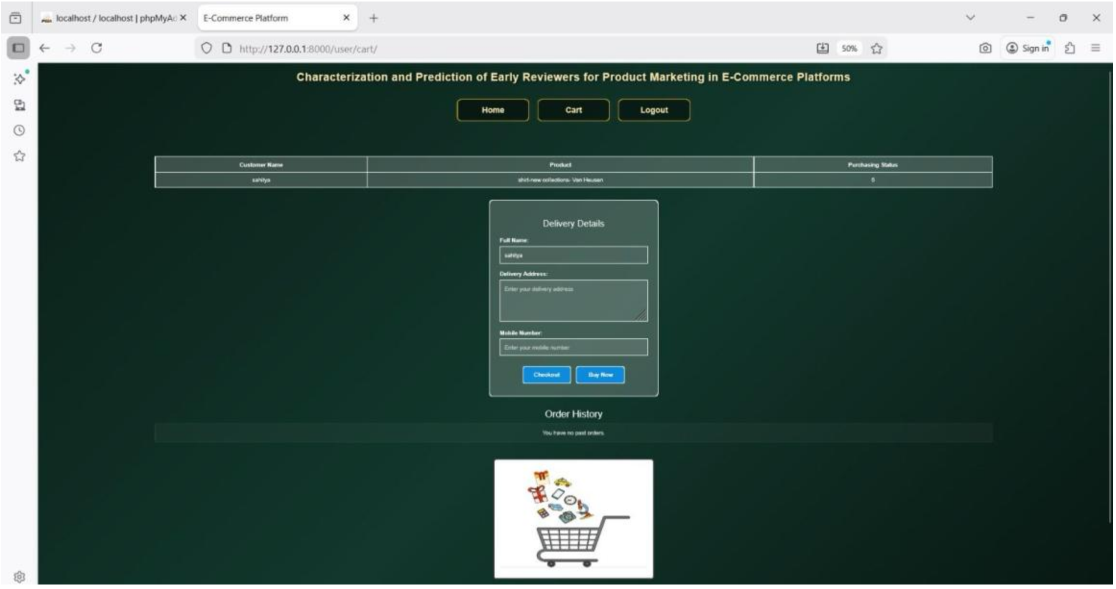
Step 7: This page displays user ratings and reviews for products.



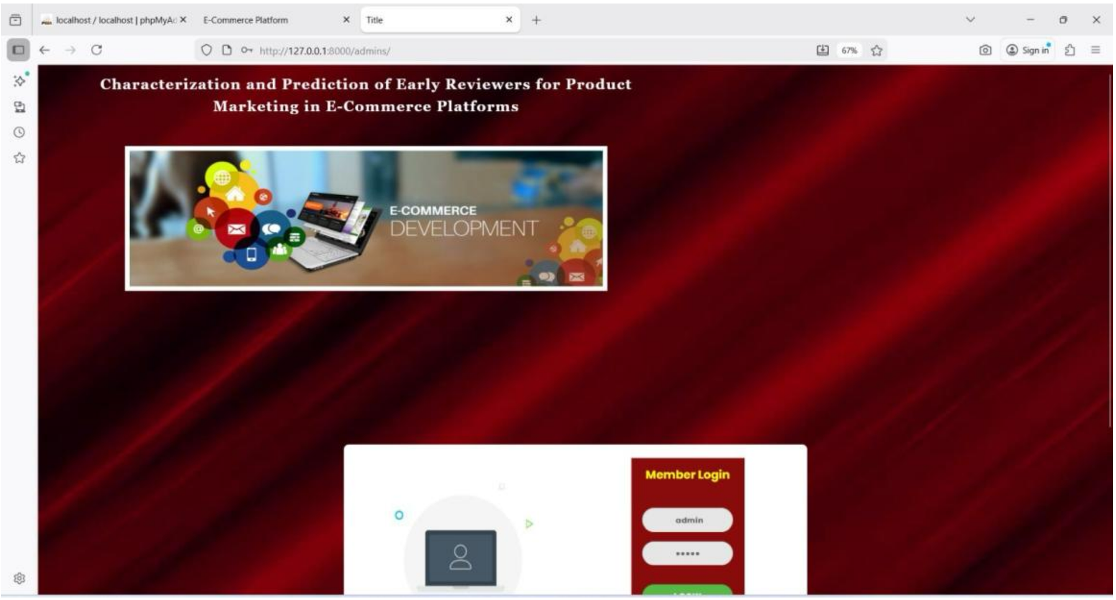
Step 8: This page allows users to give ratings and reviews for the selected product.



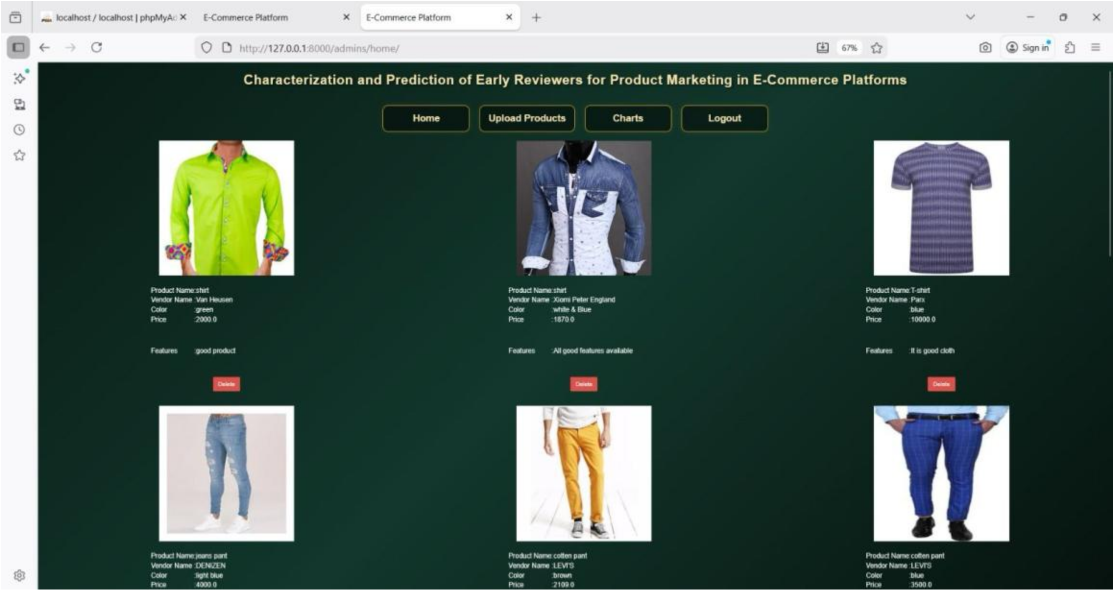
Step 9: This page displays cart details and allows users to enter delivery information and place the order.



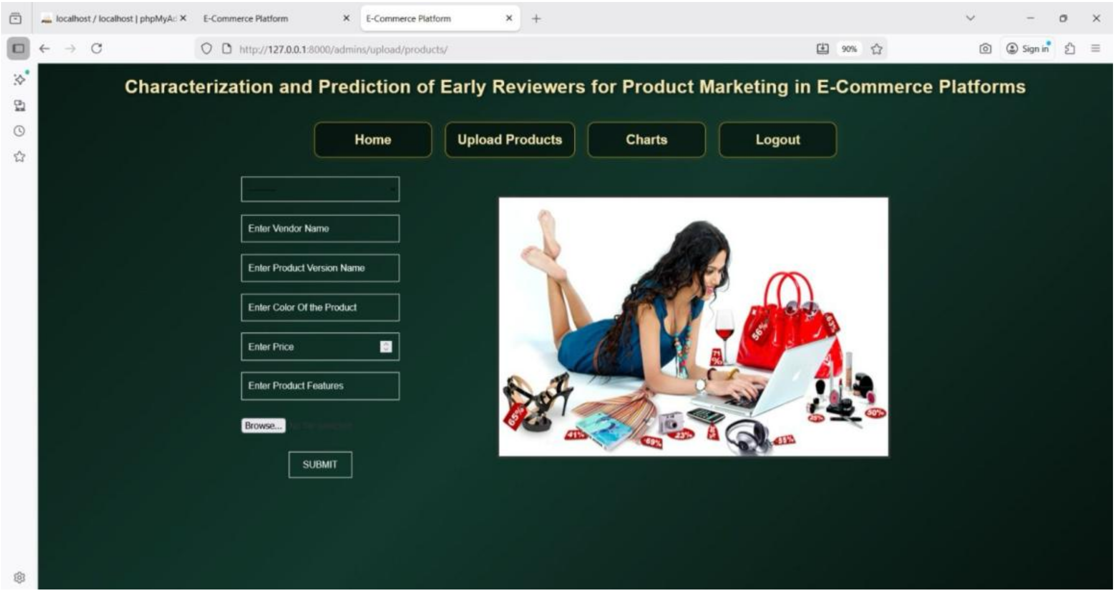
Step 10: This page allows admin to log in using admin credentials.



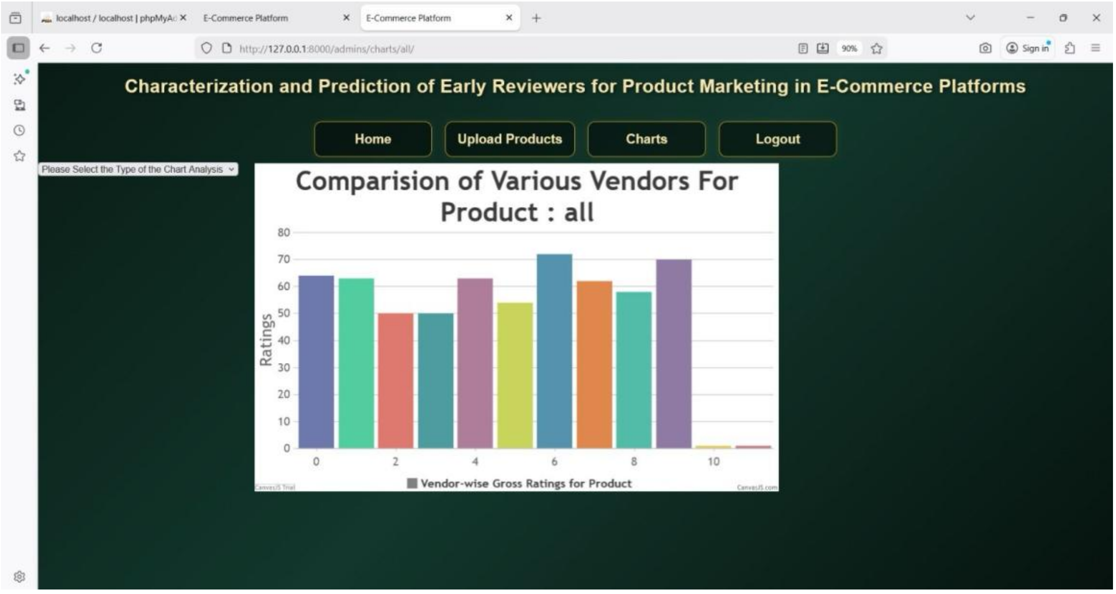
Step 11: This page displays all products and allows admin to manage them.



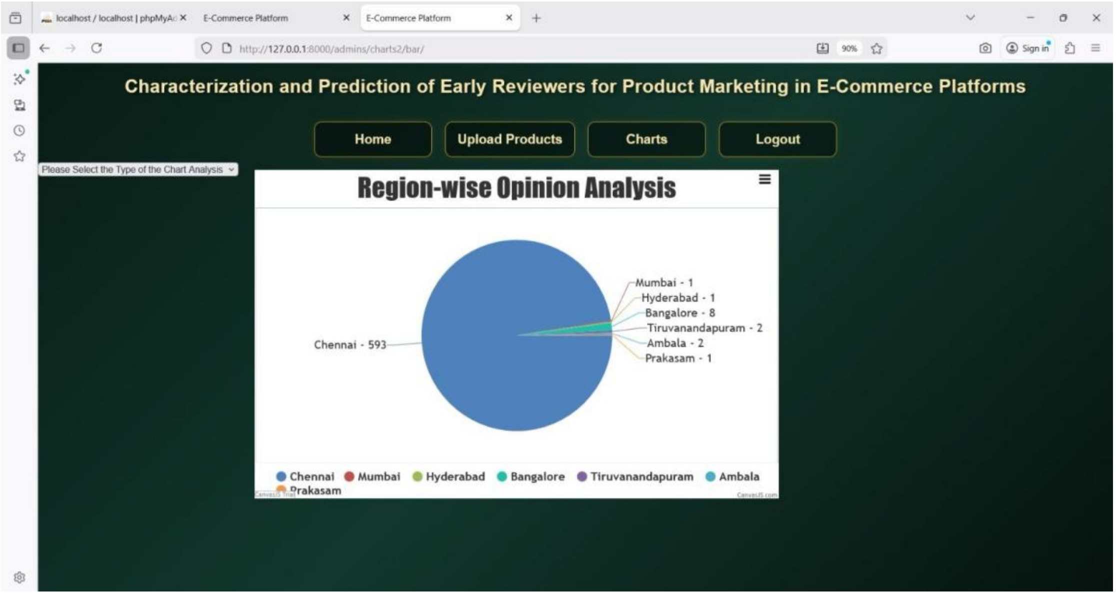
Step 12: This page allows admin to add new products to the system.



Step 13: This page shows comparison of vendors based on product ratings.



Step 14: This page shows region-wise opinion analysis using a pie chart.



8.CONCLUSION AND FUTURE SCOPE

CONCLUSION

The project **“Characterization and Prediction of Early Reviewers for Product Marketing in E-Commerce Platforms”** has been successfully developed and implemented. The system provides a complete e-commerce environment where users can register, login, browse products, add items to cart, place orders, and submit ratings and reviews, making it user-friendly and efficient.

The main objective of the project is to analyze the impact of early reviewers on product marketing. The system performs sentiment analysis on user reviews and classifies them as positive, negative, or neutral. It is observed that early reviewers tend to give higher ratings and more helpful feedback, which influences product popularity and helps in understanding customer behavior.

The admin module enables effective management of products and provides graphical analysis through charts, making it easier to interpret user feedback and trends. Overall, the system integrates user interaction, product management, and data analysis effectively, providing valuable insights for real-world e-commerce applications.

FUTURE SCOPE

In future, the system can be enhanced by integrating advanced machine learning algorithms to improve the accuracy of sentiment prediction. It can also be extended to support real-time data analysis and personalized recommendation systems for better user experience. Additionally, developing a mobile application and improving scalability can make the system more efficient and suitable for real-world applications.

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